Department of Mathematics
Franklin College of Arts and Sciences
UNIVERSITY OF GEORGIA

"Education costs money. But then so does ignorance." Sir Claus Moser

CALCULUS II for Science and Engineering, MATH 2260 INTENSIVE SECTION
SYLLABUS Spring 2026
Dr. Townsley

LECTURE: CRN 41470 MWF 8:15-9:35AM and Thursday 8:40-9:35AM in Boyd 204

OFFICE: BOYD 320 **email** Townsley@math.uga.edu
homework website: We use WebAssign, accessed through ELC

Drop-In OFFICE HOURS: MWF 11:30am-1:00pm, MW 2:45-3:45pm, Thur 10:00am-Noon. Dr. Townsley is not available on Tuesdays. You may email for other appointment times, including by Zoom

TEXT: We use an open-source text, delivered by OpenStax. Link is provided through ELC, and for vol 2: https://openstax.org/details/books/calculus-volume-2

COURSE DESCRIPTION: This course is a continuation of Calculus I, with focus on the applications of the definite integral. Additional topics include methods of integration, sequences and series, convergence tests, power series and Taylor Series, and vectors in 3-space. Webassign is the online homework server. This (intensive) section has extra time built into each week at no extra cost.

COURSE GOALS: A primary goal is to develop understanding of the mathematical content listed below, and how to use it to solve practical applications. Additional goals include the continued development of reasoning and problem solving skills. These goals may be achieved by preparing for and participating in daily lectures and discussion and working on assigned homework/quizzes/exams. Finally, you develop your ability to work together with colleagues and develop communication skills through challenge projects (outside of class) and other group assignments (in class).

TOPICAL OUTLINE (Vol 2 Sections 1.3-1.7, 2.1-2.4, 3.6, 3.1-3.5, 4.1, 4.3, 3.7, 5.1-5.6, 6.1-6.4, Vol 3 Sections 11.1-11.5 with review of Vol 1 4.8 as needed)

1. Review of the definite integral, Fundamental Theorem of Calculus, and Areas between Curves.
2. Applications of the definite integral: volumes of solids by slices and revolution, arclength and surface area, work.
3. Numeric Integration.
4. Techniques of integration: integration by substitution, integration by parts, trigonometric integrals, trigonometric substitution, partial fractions.
5. Solution of ordinary differential equations by separation of variables and initial value problems.
6. Convergence of improper integrals by direct computation and by comparison tests.
7. Sequences and series, definition of convergence/divergence and tests for convergence.
8. Power series, Taylor series, convergence set, and convergence theorems.
9. Introduction to vectors in 3-space, dot and cross products, lines and planes in space.

The Family Educational Rights and Privacy Act of 1974 opens many student records for the student's inspection. Graded group work is accessed by all group members; individual students work is viewed only by student and professor.

REQUIREMENTS: There is a prerequisite of Math 2250 or placement. Professional courtesy toward your instructor and your classmates is expected. There are no makeups for missed in-class exams and quizzes, see below for generous grading policy. In the case of documented illness, a makeup final exam is possible.

ATTENDANCE and CLASSROOM POLICY: Students are expected to attend classes regularly. A student who incurs an excessive number of absences may be withdrawn from a class at the discretion of the professor. In this class, we interpret "excessive" to mean four or more unexcused absences. In this course, ***you will be treated with respect***, and I welcome students regardless of visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. To preserve a productive learning environment for your colleagues, you are expected to be on time and remain in class for the entire period. Excessively late students are not admitted to class.

TEST DATES: FOUR exams will be given. The **tentative** schedule is Mon Feb 2, Wed Feb 23, Fri April 3, Wed April 22. Since this term begins a new course calendar, those dates are **TENTATIVE**. There will be (at least) 10 quizzes announced in eLC. There may be additional (announced or pop) quizzes. The comprehensive final exam is (41457) Wed Apr 29, 8:00-11:00am; Monday May 4 8:00-11:00am (41470) and Noon-3:00pm (40638).

TECHNOLOGY: Use of calculators is encouraged. Specifically, I recommend the use of the TI 30XS Multiview OR LOWER. Some exams/quizzes will be entirely calculator-free. Others may utilize calculators without graphics, solution operations (no TI 36), or symbolic algebra capability (no TI-83/84, 89, 92, NSpire, or other CAS). At all quizzes and exams, the calculator memory must be free of additional mathematical material. Dr. Townsley has a FEW calculators to loan for tests/quizzes.

GRADING: A total of 600 points are possible. Each midterm exam is worth 100 points, but only the best 3 scores will contribute to your final grade. The final exam is 150 points. Quizzes are worth 15 points each: your best 7 scores count for a total possible 105 points. There may be extra announced or unannounced quizzes, which improve your quiz grade (more scores dropped). Homework is normalized to a percentage of all assigned problems, and contributes 45 points to your grade (% * 45 is how it is computed, but in fact you don't have to complete every problem). Take your total earned points and divide by 6 to get a 100% value: The following grading scale will apply: ≥ 91 = A, 89-90 = A -, 87-88 = B+, 81-86 = B, 79-80 = B -, 77-78= C+, 71-76 = C, 69-70 = C -, 60-68 = D, and below 60 is an F. Dr. Townsley does NOT round. There are extra credit opportunities available during the term (test corrections, challenge presentations, etc) and students can earn up to 6 extra credit points (1% of the total grade) this way.

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary. It is the responsibility of the student to seek clarification of the grading policy and/or course requirements and procedures from the instructor. Handouts in this class, including exams, are copyrighted materials. Unauthorized use of copyrighted materials, including posting them on the web, is prohibited.

ELECTRONIC DEVICES POLICY: One aspect of being a member of a community of scholars is to show respect for others by the way you behave. One way of showing respect for others in the educational community is to do your part to create or maintain an environment that is conducive to learning. Allowing your cell phone to chime or vibrate in class is inappropriate because it disrupts the classroom experience for your classmates. For the sake of your classmates, *you are expected to turn off your cell phone or set it to mute/silence BEFORE you enter class—every class*. Furthermore, if you use your cell phone *in any inappropriate manner* during class (e.g. text messaging, games, social media, etc.), you will be dismissed from class and will forfeit any points you might have earned in that class period. If you use your cell phone

in any manner during a test or quiz, you have violated the academic honesty policy. (This policy also applies to LAPTOPS, pagers, tablets, smart watches, PDAs, Treos, MP3 players and all other electronic communication and/or data storage devices.) There will be times in class when you may wish to snap a pic of work, or use a device to call up homework problems—Dr. Townsley will suggest times when this is acceptable. Using your laptop or tablet for notetaking is fine, provided you do not violate the above policy—a violation by any student loses this privilege for the entire class.

ACCOMMODATION FOR DISABILITIES

If you plan to request accommodations for a disability, please register with the *Accessibility & Testing*. They can be reached by visiting Clark Howell Hall, calling 706-542-8719 (voice) or 706-542-8778 (TTY), or by visiting https://accessibility.uga.edu/. It is important to do this early, to ensure your ability to participate fully in this class.

Tech support

eLC Student Support : https://help.elc.uga.edu/contact/
WebAssign Student Support: https://webassign.com/support/student-support

ACADEMIC HONESTY POLICY: All academic work must meet the standards contained in *A Culture of Honesty*. (https://honesty.uga.edu and you can "take the pledge" and read the document). Students are responsible for informing themselves about these standards before performing any academic work. This policy defends the academic integrity of all student work, and will be uniformly applied to all students in the class. Student work in this class is individual, without notes, with the exception of some quizzes which may allow notes and may be a group assignment.

UGA Student Honor Code: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." UGA's policy and procedures for handling cases of suspected dishonesty can be found at the same website. Generative AI tools can be useful as study aids and for particularly knotty homework problems. For this course, generative AI tools may only be used if explicitly permitted, with clear disclosure of their use in submissions. Other than homework assignments and personal studying, AI tools should be not accessed. All work must reflect your own understanding, and misuse of AI tools will be addressed according to UGA's academic integrity policies.

REMARKS: Questions are encouraged at all times. Scheduling difficulties can be amicably settled by PRIOR discussion. Please contribute as a positive member of this learning community. Note that there is supplemental instruction for this course, UNIV 2260. Notice that there is no school on the dates Jan 19, Mar 9-13. The last day to drop any course is April 9.

STUDENT SUCCESS RESOURCES

The Office for Student Success and Achievement (OSSA) empowers students to achieve success throughout their academic journey. Through free peer tutoring, academic coaching, UNIV student success courses, Bulldog Basics, and first-generation student support, we promote well-being, student learning, and community building. To connect with OSSA, email *ossa@uga.edu*, call (706) 542-0163, or visit Milledge Hall (near Reed Hall and Sanford Stadium).

UGA WELL-BEING RESOURCES

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community.
Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.
UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- Well-being Resources: well-being.uga.edu
- Student Care and Outreach: sco.uga.edu
- University Health Center: healthcenter.uga.edu
- Counseling and Psychiatric Services: caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273
- Health Promotion/ Fontaine Center: healthpromotion.uga.edu
- Accessibility & Testing: accessibility.uga.edu

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting well-being.uga.edu. One stop shopping: for health and wellness resources, free tutoring, stuff swap, food pantry, lots of cool links, see https://instruction.uga.edu/resources/student/

DISCLAIMER
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